Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *AFRIDOC_MT* | ***Context:*** *Ethiopia Ministry of Health — Amharic Clinical Translation Deployment*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Both benchmark and deployment are text-only Amharic in Ethiopic/Ge'ez script; this is a strong form match. The benchmark explicitly handles Amharic's high token counts due to non-Latin script and increased the Amharic token limit during evaluation [Q47, Q127, Q128, Q161, Q162]. Document-level structure is preserved with NLTK + linguist sentence segmentation [Q19–Q21] and pseudo-document chunking (k=10) [Q48, Q49] to fit model context limits. Dataset inspection confirms full Ethiopic-script Amharic documents are present with multi-paragraph structure [DATASET-D1, D3]. Minor concerns: full-document translation is infeasible for most open LLMs given 8,192-token context [Q88, Q89], imposing a structural ceiling; the benchmark uses CSV with paragraph separator conventions [Q118, Q119, Q122] which may differ from MOH document formats. With IF marked LOWER priority by the user (no modality/script mismatch), these concerns do not warrant a lower score.

ID	Evidence
Q19	“While our corpus is initially structured at the article level, we aim to make it suitable for sentence-level translation tasks as well.” (p.3)
Q20	“To achieve this, we segmented the raw articles into sentences using NLTK (Bird et al., 2009).” (p.3)
Q21	“To ensure high segmentation quality, we recruited a linguist and a professional translator to verify the correctness of the segmentation and make corrections as needed.” (p.3)
Q47	“An initial analysis showed that some models were unable to process entire documents due to input length limits, which were exceeded by token counts in some languages (Amharic and Yorùbá).” (p.5)
Q48	“To address this, we adopted a similar approach to Lee et al. (2022), splitting documents into fixed-size chunks of k sentences to fit within token limits; the final chunk may contain fewer than k sentences.” (p.5)
Q49	“To select an appropriate chunk size, we conducted initial tests with k = 1 (sentence-level), 5, 10, and 25, choosing k = 10 for our experiments.” (p.5)
Q88	“Except for LLama3.1, all other open LLMs have a context length of 8192 tokens, while encoder-decoder models were primarily based on T5.” (p.9)
Q89	“This makes it difficult to use the context length beyond a certain limit, making full document translation infeasible.” (p.9)
Q118	“Each file contains 2500 sentences, and they are named in the format of a serial number followed by your first name.” (p.18)
Q119	“Please do not delete double empty rows, as they serve to separate paragraphs. Also, avoid deleting any rows, columns, or provided text.” (p.18)
Q122	“The files are in .csv format, and you can open them using Google Sheets or Microsoft Excel (for offline work).” (p.18)
Q127	“Our analysis revealed that Amharic and Yorùbá—languages with unique characteristics such as non-Latin scripts and diacritics, respectively—had the largest average token counts across the tokenizers.” (p.19)
Q128	“To accommodate both languages in our experiments, we chose pseudo-documents with k=10.” (p.19)
Q161	“However, for Amharic, when translating pseudo-documents with 25 sentences and full documents, there were instances exceeding the 95th percentile derived from the training statistics.” (p.21)
Q162	“Therefore, we increased the token limit specifically for Amharic.” (p.21)
D1	የሞቃታማ ዐውሎ ነፋሶች ... ለሞቃታማ አውሎ ነፋሶች፣ እንዲሁም ሀይለኛ አውሎ ነፋስ ወይም አውሎ ነፋሶች ... ኃይለኛ ክብ አውሎ ነፋሶች ... 233 000 ሰዎችን ገድለዋል — Full Amharic document on tropical cyclones (WHO article), multi-sentence, Ethiopic script, with clinical/epidemiological content and specific numbers

D3	አካል ጉዳተኝነት ሰው የመሆን አካል ነው። ሁሉም ሰው ማለት ይቻላል በሕይወታቸው ላይ በሆነ ወቅት ጊዜያዊ ወይም ዘላቂ የአካል ጉዳት ያጋጥማቸዋል — Amharic WHO article on disability — prose narrative, full document preserved

Strengths

- Amharic in Ethiopic/Ge'ez script is natively present and structurally accommodated, including increased token limits to handle Amharic's high token counts [Q127, Q128, Q162]
- Document-level structure is preserved with linguist-verified sentence segmentation [Q20, Q21]
- Pseudo-document chunking enables evaluation under realistic context-length constraints [Q48, Q49, DATASET-D1]
- Multi-parallel structure provides English↔Amharic source-target rows directly usable for the deployment direction [DATASET-D12]

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D12	"OCV ni zana ambayo hutumiwa pamoja na hatua za kudhibiti kipindupindu." (sw) / "OCV jẹ́ ohun èlò fún àfikún àwọn ìlànà fún ṣíṣe àmójútó àrùn onígbá méjì" (yo) — Sentence-level multi-language row — all six languages present including Amharic

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution matches: text in Ethiopic script for Amharic. The benchmark documents that Amharic and Yorùbá produce the largest tokenizer counts and adjusted the Amharic token limit accordingly [Q127, Q161, Q162]. Dataset inspection confirms full Ethiopic-script multi-paragraph Amharic [DATASET-D1, D3, D5].

IF-2: MOH/Addis Ababa-side infrastructure can support Unicode Ethiopic-script processing; deployment is text-only with print and digital distribution [deployment context]. End-user connectivity (19.4% internet penetration nationally [WEB-11]) is relevant for distribution but not for the MT system's input form.

IF-3: Document-level format with paragraph preservation is supported via CSV with double-empty-row paragraph separators [Q118, Q119, Q122]. However, MOH distribution materials may use formats (Word, PDF, print-ready layouts) requiring additional conversion. The benchmark's full-document translation is infeasible for most open LLMs (8,192-token ceiling) [Q88, Q89], constraining how the system handles longer MOH documents — a real but moderate constraint.

IF-4: Minor form mismatches: (1) full-document translation infeasibility for most LLMs forces pseudo-document chunking, which the benchmark itself shows can degrade African-language translation quality vs. sentence-level [Q61, Q72]; (2) CSV-with-paragraph-separator input format differs from typical MOH document formats but is a reasonable proxy for document-level structure.

ID	Evidence
Q61	“Pseudo-document translation is worse than sentence-level translation when translating into African languages Our results from pseudo-document translation show a performance drop across different models compared to sentence-level translation, especially when translating into African languages.” (p.6)
Q72	“Human evaluation results align closely with d-chrF, which favors sentence-level translations over pseudo-document translations when translating into African languages.” (p.8)
Q88	“Except for LLama3.1, all other open LLMs have a context length of 8192 tokens, while encoder-decoder models were primarily based on T5.” (p.9)
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D5	ፀረ-ተሕዋስያንን መቋቋም(AMR) ... ባክቴሪያ፣ ጥገኛ ተውሳኮች፣ ቫይረሶችና በፈንገሶች ምክንያት — Amharic AMR article — specialized medical terminology in Ethiopic script
WEB-11	Ethiopia internet penetration ~19.4% in early 2024 — affects distribution channel but not MT system input form (https://datareportal.com/reports/digital-2024-ethiopia)

Information Gaps

- Whether actual MOH source documents fit within the benchmark's pseudo-document (k=10) chunking assumption or routinely exceed it

Requires Expert Verification

- Confirmation that MOH document formats (Word/PDF/print) can be losslessly converted to the benchmark's text-with-paragraph-separator representation